

Vibrating tilt platform enhancing immersive experience in VR

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ABSTRACT

Virtual reality systems, since their initial development, suffered some disadvantages. One of them was the fact that they had only visual interfaces. This limitation, however, has been successfully overcome with the development of haptic technology. Peripheral solutions reinforcing and enriching VR experience are now commonplace, and many haptic systems are being developed for deepening VR immersion. This paper discusses a new peripheral solution – a vibrating tilt platform with three angles of inclination to be used for enhancing VR experience.

Index Terms: Virtual reality, Virtual environment, Mechanical design, Human-computer interaction, Human-centered computing, Interaction, HCI design, Usability, Tilt platform, EEG.

1 INTRODUCTION

Virtual reality systems, since their initial development, suffered some disadvantages. One of them was the fact that they had only visual interfaces. This limitation, however, has been successfully overcome with the development of haptic technology. Peripheral solutions reinforcing and enriching VR experience are now commonplace, and many haptic systems are being developed for deepening VR immersion [1, 2]. This paper discusses a new peripheral solution – a vibrating tilt platform with three angles of inclination to be used for enhancing VR experience.



Figure 1: The prototype of the tilt platform.

2 METHODS

The vibrating tilt platform (see Fig. 1) was created to provide multi-sensory stimulation of the user of a VR system. The user's participation is passive and static by design, and they are required to stand

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still. To stress their static location, there are marked areas on the plate indicating where to put their feet. The research procedure is initiated with a thorough explanation and guidance on the usage of the platform. There are two basic sensory functionalities. The first one comprises two-axis $\pm 10^\circ$ tilt and changing vertical axis ± 50 mm, all under control of a set of three electric actuators. The use of actuators allows the position of the platform to be uniquely determined. This, in turn, removes the necessity of software control over the positioning of individual actuators in relation to each other. The second functionality of the tilt platform is to induce vibrations and shocks. Four small vibration motors were applied for that purpose, and they create a sensation of “tingling” of the surface.

2.1 VR application

Game engine Unity and C# programming language were used to develop three events in the virtual reality environment. They imitate an earthquake situation, including the fracture and collapse of the simulated substrate (see Fig. 2). Each event differs in the intensity and quantity of applied haptic experiences. The first event comprises solely audiovisual stimuli. The second one is reinforced with the overload effects of the moving platform. The last one, in addition to effects applied in the second event, includes haptic effects imitating vibration of the ground surface. The order of the events is randomized and the target platform for the VR application is HTC Vive. The application can be ported to different systems with ease.



Figure 2: Sample scene from the VR experience.

2.2 Compliance with ethical standards

All objectives of the experiment were achieved with high ethics level, according to the Research Ethics Committee of the Lodz University of Technology.

2.3 Assessment procedure

The platform was evaluated using objective (brain activity data acquired with an EEG sensor) and subjective (questionnaire) features.

3 EXPERIMENT PROCEDURE AND RESULTS

22 volunteers took part in the study. The first stage of the experiment was a pre-test filled in by the subjects in which they provided information about the subjective assessment of their psycho-physical state. The VR session using the tilt platform was the next stage of the experiment. When it was finished, the volunteers were asked to

fill in a post-test to provide information about the changes in their psycho-physical state and some feedback on the usability aspects of the VR experience they had just taken part in.

The participants were characterized as follows:

On average, samples were 28.4 years old (median = 23, std = 8.8, range = 21-48) and the group comprised twelve men (mean = 25.0, median = 23.0, std = 5.92, range = 21-38) and ten women (mean = 32.4, median = 23.0, std = 9.83, range = 21-48). Most participants were familiar with VR technology and had experienced it at least few times before the session (64%). Only 9% of the users had not before tried any VR experience.

The comparison of pre- and post-tests indicated that the subjective assessment of participants' psycho-physical state changed slightly after the VR earthquake experience. Overall, the changes were minor - around 1-point in the scale. 63.6% volunteers claimed to be more energetic and stimulated after VR experience. Surprisingly, 45% evaluated their mood as more positive in post-test even though the real earthquake experience is bound not to produce a similar effect. The volunteers, on the whole, did not feel more tense or relaxed. The usability questionnaire revealed that the deepest immersion in the VR experience was achieved in the scenario with additional platform tilt (average = 3.5).

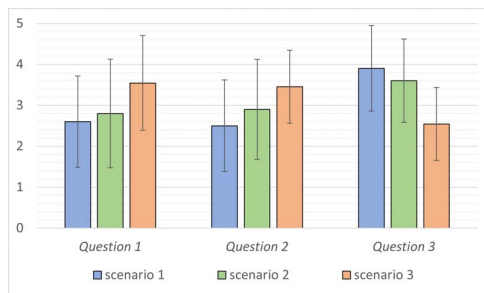


Figure 3: Results of usability testing.

3.1 Electroencephalography analysis

Figure 4 presents an example of brainwaves recordings interpreted by Looxid Link as two states (attention and relaxation) of a 33-year-old woman during the session divided into separate scenarios.

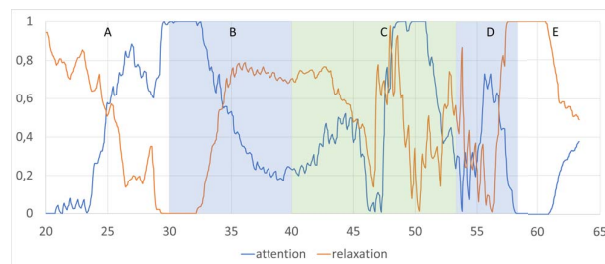


Figure 4: Brainwaves recordings interpreted by Looxid Link into two stages (attention and relaxation) of a 33-year-old woman during the whole session divided into three (B, C, D) separate scenarios.

4 CONCLUSION

This paper presents the vibrating tilt platform, which reinforces the realism of the VR experience simulating earthquakes. This platform can be easily adjusted to be used for rehabilitation for motor disabilities or phobias, rescue environments, and educational issues. Both objective (EEG) and subjective (questionnaire) data

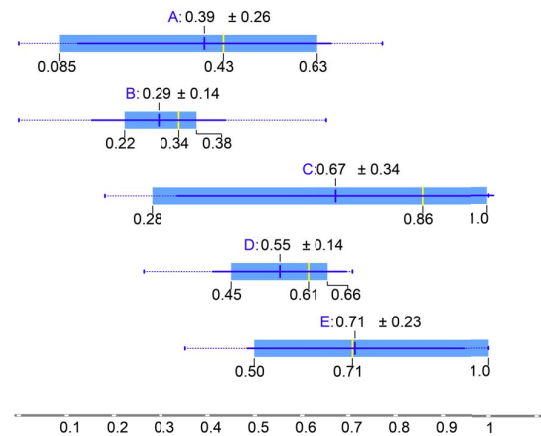


Figure 5: Boxplot presenting the level of relax averaged over all participants during the whole session divided into three (B, C, D) separate scenarios.

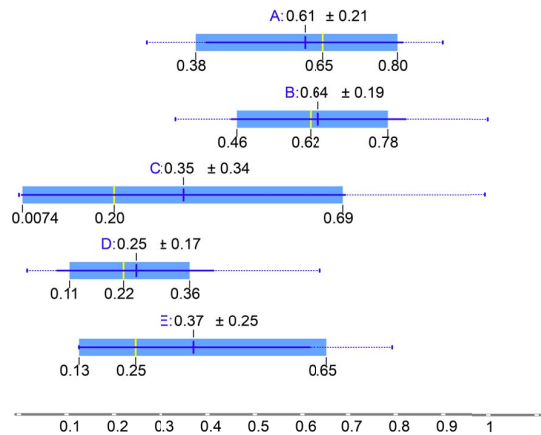


Figure 6: Boxplot presenting the level of relaxation averaged over all participants during the whole session divided into three (B, C, D) separate scenarios.

indicates clearly that the haptic sensations induced by the SIM-Motion platform increase the users' level of immersion in the VR environment.

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